



COURSE SYLLABUS						
Cours e Code	COMP 008	Course Title	Data Communications and Networking		Course Credit	3
Course Description		This course introduces the principles of data communications and computer networking, focusing on network architectures, protocols, and transmission methods based on the updated OSI and TCP/IP models. It emphasizes cloud computing integration, Virtual Private Clouds (VPC), and Software-Defined Networking (SDN). Students will gain practical skills in designing, configuring, and securing networks through hands-on activities and simulations.				
Pre-Requisites			Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				





**INSTITUTIONAL
LEARNING OUTCOMES
(ILO)**

- As a polytechnic state university, PUP shall develop its students to possess:
1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
 2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
 3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
 4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
 5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.





	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10.Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11.Produce technologies for commercialization of livelihood improvement.

PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO				
	Apply knowledge of computing, science, and mathematics appropriate to th				

QS WORLD UNIVERSITY RANKINGS

WURI

Rank 21
GENERATIVE AI
APPLICATION

Rank 39
STUDENT SUPPORT
ENGAGEMENT

Rank 50
SYMBOL/
PROMOTION

2025

7th in the Philippines
92nd in Southeast Asia
541-560 in Asia

The WORLD UNIVERSITY RANKINGS for INNOVATION

QS STARS

Times Higher Education

5+ stars

Employability
Online Learning
Internationalization

4+ stars

Teaching
Academic Development
Inclusiveness

3+ stars

Social Responsibility
Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

Impact Rankings 2024

DEBUTS AT 1501+

o ILOs			
5	7	8	9



	Analyze complex problems, and identify and define the computing requirements appropriate to its solution												
	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.												
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.												

2025

7th in the Philippines
92nd in Southeast Asia
541-560 in Asia

The WORLD UNIVERSITY RANKINGS for INNOVATION

Rank 21
GENERATIVE AI APPLICATION

Rank 39
STUDENT SUPPORT ENGAGEMENT

Rank 50
SYMBOL/ PROMOTION

5+ stars

- Employability
- Online Learning
- Internationalization

4+ stars

- Teaching
- Academic Development
- Social Responsibility

3+ stars

- Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BEND)

Impact Rankings 2024

DEBUTS AT 1501+



	Integrate IT-based solutions into the user environment effectively.									
	Assist in the creation of an effective IT project plan.									
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession									
	Understand best practices and standards and their applications.									
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.									
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.									
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.									
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.									
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development									

COURSE LEARNING OUTCOMES (CLO)	Explain the fundamental concepts, components, and functions of data communication systems.	Alignment to PLOs							
		1	2	3	4	5	6	7	8

QS WORLD UNIVERSITY RANKINGS

2025
7th in the Philippines
92nd in Southeast Asia
541-560 in Asia

WURI

The WORLD UNIVERSITY RANKINGS for INNOVATION

Rank 21
GENERATIVE AI APPLICATION

Rank 39
STUDENT SUPPORT ENGAGEMENT

Rank 50
SYMBOL/ PROMOTION

QS STARS

5+ stars
- Employability
- Online Learning
- Internationalization

4+ stars
- Teaching
- Academic Development
- Inclusiveness

3+ stars
- Social Responsibility
- Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

Times Higher Education
Impact Rankings 2024
DEBUTS AT 1501+



	Differentiate and analyze various network topologies, transmission modes, and communication models.	X	X		X			
	Describe and apply the updated OSI and TCP/IP models in real-world network operations.	X	X		X			
	Demonstrate proficiency in configuring and troubleshooting basic network communication using Cisco Packet Tracer.		X	X	X	X		
	Design and implement subnetting, IP addressing, and routing schemes suitable for cloud and on-premise networks.		X	X	X			
	Apply security, monitoring, and performance optimization techniques in physical and cloud-based networks.		X		X	X		
	Integrate cloud networking concepts, including virtualization and Virtual Private Cloud (VPC), into hybrid network architectures.		X	X	X	X	X	
	Develop and present a comprehensive network design project demonstrating communication and cloud technologies.		X	X	X	X	X	X





OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)	Face-to-Face	FLTAs	1	2	3	4
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP - Gain an overview of the course subject and requirements	- Orientation about PUP Vision, Mission, and 10 Pillars - Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.		Formative Assessment (Sharing of Learning)		X			
2	Differentiate analog/digital signals and transmission modes.	Analog vs. Digital; Simplex, Half, Full Duplex.	Demonstration and discussion	Create infographic and peer-review online.	Short quiz.	Infographic submission.	X	X		
3	Identify and illustrate network topologies.	Star, Bus, Ring, Mesh, Hybrid.	Topology drawing; classroom discussion	Packet Tracer simulation practice.	Drawing check.	Simulation		X		
4	Explain OSI model layers and functions.	7-layer OSI Model (updated virtualization roles).	Layer walkthrough; sample data path.	Watch OSI in Cloud video; mapping activity.			X	X		





5	Compare OSI and TCP/IP models.	Mapping, differences, cloud stack relation.	Group mapping activity.	Collaborative activities online.	Group presentation	Slide upload.	X	X		
6	Apply error detection and correction concepts.	Parity, CRC, Hamming, Checksum.	Problem-solving drills.	Online quiz practice.	Seatwork check	Auto quiz results.		X	X	





7	Analyze switching and routing techniques	Circuit, Packet, Message switching; routing basics.	Lab demo using Packet Tracer.	Cloud routing simulation video.	Lab report.	Reflection answers.		X		X
8	Describe Cloud computing and virtualization.	Cloud models: IaaS, PaaS, SaaS; hypervisors; containers.	Lecture-discussion	Online reading; short quiz.	Quiz.	Quiz result.		X	X	
9	MIDTERM WRITTEN EXAMINATION									
10	Explain VPC architecture and components.	Subnets, route tables, gateways, security groups.	Diagram-based lecture	AWS VPC lab simulation	In-class quiz.	VPC diagram submission.	X			X
11	Apply IP addressing and subnetting	IPv4, IPv6, CIDR, subnet planning.						X		X
12	Identify cloud network security mechanisms.	Firewalls, VPN, encryption, IAM, shared responsibility.	Computation exercises.	Interactive subnetting tool practice.	Timed exercise.	Worksheet upload.		X		X
13	Explain wireless and IoT networking.	Wi-Fi, 5G, Bluetooth, IoT protocols.	Case discussion.	Watch video; reflection writing.	Quiz.	Reflection upload.			X	X
14	Discuss hybrid and edge networking concepts.	Hybrid networks, SD-WAN, VPC Peering, Edge computing.	Demo devices and standards.	IoT video and discussion forum.	Quiz.	Forum	X		X	
15	Design a hybrid network integrating VPC.	LAN/WAN with Cloud, security and redundancy.	Case study and discussion	Research critique via LMS.					X	X





16	Evaluate network monitoring and scalability.	CloudWatch, Azure Monitor, metrics and fault tolerance.	Demo monitoring tools.	Virtual lab: run metrics.	Lab check.	Monitoring report upload.			X	X
17	Integrate all DCN and cloud concepts in final project.	Final Presentation.	Final presentation & feedback.	Submit final documentation.	Final Presentation	Project upload	X	X	X	X
18	FINAL WRITTEN EXAMINATION									

2025

7th in the Philippines
92nd in Southeast Asia
541-560 in Asia

The WORLD UNIVERSITY RANKINGS for INNOVATION

Rank 21
GENERATIVE AI APPLICATION

Rank 39
STUDENT SUPPORT ENGAGEMENT

Rank 50
SYMBOL PROMOTION

5+ stars

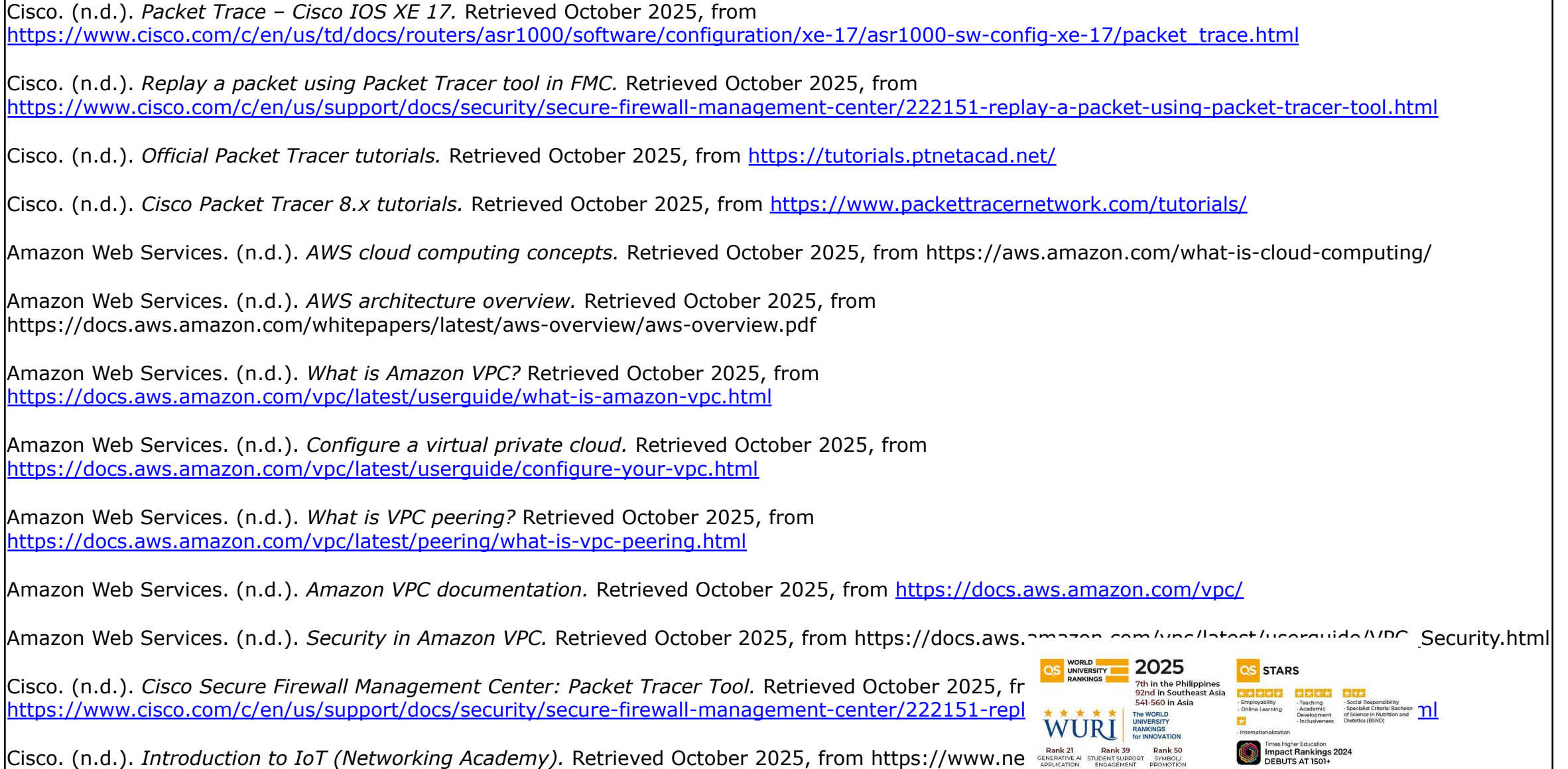
- Employability
- Online Learning
- Internationalization
- Teaching
- Academic Development
- Social Responsibility
- Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

Impact Rankings 2024

DEBUTS AT 1501+



		Resources and Applying Strategies in Teaching Prose Competency 4: Recognizing Assessment in Teaching Prose	achievement which are consistent with the selected competencies • Learning Plan Making according to the English curricula that develops higher order thinking skills of learners through the use of prose and dramatic texts		• Quizzes					
18	FINAL EXAMINATION									
LIST OF REFERENCES BOOKS: WEBSITE: Cisco Networking Academy. (n.d.). <i>What is Cisco Packet Tracer?</i> Retrieved October 2025, from https://www.netacad.com/cisco-packet-tracer Cisco Learning Network. (n.d.). <i>Packet Tracer and alternative lab solutions.</i> Retrieved October 2025, from https://learningnetwork.cisco.com/s/packet-tracer-alternative-lab-solutions PacketTracerNetwork. (n.d.). <i>Download & install Cisco Packet Tracer.</i> Retrieved October 2025, from https://www.packettracernetwork.com/download-packet-tracer.html Sulaiman, M. K. E. B. (2020). <i>Teaching creativity using a realistic multi-user operation: Packet Tracer</i> [Preprint]. arXiv. https://arxiv.org/abs/2101.02009 Cisco. (n.d.). <i>Packet-tracer explained for Nexus 9000 series.</i> Retrieved October 2025, from https://www.cisco.com/c/en/us/support/docs/switches/nexus-9000-series-switches/210592-Nexus-900										
WORLD UNIVERSITY RANKINGS 2025 7th in the Philippines 92nd in Southeast Asia 541-560 in Asia WURI The WORLD UNIVERSITY RANKINGS for INNOVATION Rank 21 GENERATIVE AI APPLICATION Rank 39 STUDENT SUPPORT ENGAGEMENT Rank 50 SYMBOL/ PROMOTION QS STARS • Employment • Online Learning • Internationalization • Teaching • Academic Development • Social Responsibility • Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND) Times Higher Education Impact Rankings 2024 DEBUTS AT 1501+										





Cisco. (n.d.). *Wireless networking basics*. Retrieved October 2025, from <https://www.cisco.com/c/en/us/solutions/enterprise-networks/wireless/index.html>

Amazon Web Services. (n.d.). *Network-to-Amazon VPC connectivity options (Whitepaper)*. Retrieved October 2025, from <https://docs.aws.amazon.com/whitepapers/latest/aws-vpc-connectivity-options/introduction.html>

Amazon Web Services. (n.d.). *Amazon VPC Lattice features*. Retrieved October 2025, from <https://aws.amazon.com/vpc/features/>

Amazon Web Services. (n.d.). *Amazon CloudWatch user guide*. Retrieved October 2025, from <https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/WhatIsCloudWatch.html>

Amazon Web Services. (n.d.). *Monitoring Amazon VPC network traffic*. Retrieved October 2025, from <https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs.html>

Cisco Networking Academy. (n.d.). *Networking Essentials course materials*. Retrieved October 2025, from <https://www.netacad.com/courses/networking-essentials>

Amazon Web Services. (n.d.). *AWS Educate Cloud Labs*. Retrieved October 2025, from <https://aws.amazon.com/education/awseducate/>

CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING

2025
7th in the Philippines
92nd in Southeast Asia
541-560 in Asia

The WORLD UNIVERSITY RANKINGS for INNOVATION

Rank 21
GENERATIVE AI APPLICATION

Rank 39
STUDENT SUPPORT ENGAGEMENT

Rank 50
SYMBOL/ PROMOTION

QS STARS

Employability

Teaching

Academic Development

Internationalization

Social Responsibility

Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

Times Higher Education
Impact Rankings 2024
DEBUTS AT 1501+



Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.

Students' Responsibilities

Students are expected to:

- Take time to read and watch the provided class learning materials.
- Fulfill class requirements such as posting on discussion boards and submitting group presentations.
- Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.





POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE VICE PRESIDENT FOR CAMPUSES
TAGUIG CAMPUS
Bachelor of Science in Information Technology
(BSIT)





2025
7th in the Philippines
92nd in Southeast Asia
541-560 in Asia



The WORLD
UNIVERSITY
RANKINGS
for INNOVATION

Rank 21
GENERATIVE AI
APPLICATION

Rank 39
STUDENT SUPPORT
ENGAGEMENT

Rank 50
SYMBOL/
PROMOTION



STARS

★★★★★
- Employability
- Online Learning
- Internationalization

★★★★★
- Teaching
- Academic
Development
- Inclusiveness

★★★★★
- Social Responsibility
- Specialist Criteria: Bachelor
of Science in Nutrition and
Dietetics (BSND)



Times Higher Education
Impact Rankings 2024
DEBUTS AT 1501+

General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: <https://bit.ly/PUPSINTA>

THE COUNTRY'S 1st POLYTECHNICU



The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.

We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.

Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.

Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.

Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.

Class announcements will be communicated through the class page or posted through university-issued (i.e., official) email accounts. Students should check their email regularly.

Intellectual Property and Copyright.

Original materials created for this online class is covered by copyright. If you use any (third-party) instructional materials (e.g., YouTube videos) might be subject to copyright while others (e.g., some assigned readings) might be subject to fair use.



Always be honest.

Some materials will be available for download.



	be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a
--	---





Mid Term		Final Term	
Class Standing	70%	Class Standing	70%
Exercises/Activities		Exercises/Activities	
Summative Quizzes		Summative Quizzes	
Recitation and Attendance		Recitation and Attendance	
Reflection Paper		Reflection Paper	
Midterm Examinations	<u>30%</u>	Final Examinations	
	100%		30%
			100%

	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.		
COURSE REQUIREMENT/S	GRADING SYSTEM		
Students are expected to actively participate in both face-to-face and online class discussions, complete all assigned readings, quizzes, and laboratory exercises using Cisco Packet Tracer, and submit all practical activities, network design outputs, and written reports within the given deadlines. Each student must also take the midterm and final examinations, which include both theoretical and practical components.	Final Grade = (Midterm Grade + Final Term Grade) / 2		





CLASS INFORMATION	FACULTY INFORMATION
Course, Year and Section: BSIT 3-1	Name of Faculty: Christian Jim Fajardo Minalabag
Day and Time: T/T 01:00PM -04:00PM/04:00PM-06:00PM	Consultation Time:
Room: DOST Laboratory	Office Tel. No./ Mobile Phone No.
School Year: 2025=2026 Semester: 1st	E-Mail Address:

Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
Faculty Member	MS. DIANNE S. SEGURORA Head of Academic Programs	DR. RHYAN V. MOLINAR Campus Director	DR. ARMANDO A. TORRES Vice President for Campuses



2025

7th in the Philippines
92nd in Southeast Asia
541-560 in Asia

The WORLD
UNIVERSITY
RANKINGS
for INNOVATION

Rank 21
GENERATIVE AI
APPLICATION

Rank 39
STUDENT SUPPORT
ENGAGEMENT

Rank 50
SYMBOL/
PROMOTION



5+ stars
Employability
Online Learning
Teaching
Academic
Development
Inclusiveness
Social Responsibility
Specialist Criteria: Bachelor
of Science in Nutrition and
Dietetics (BSND)

